



PROGRAMME OF
THE EUROPEAN UNION



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#EUSpace

The Copernicus Health Hub

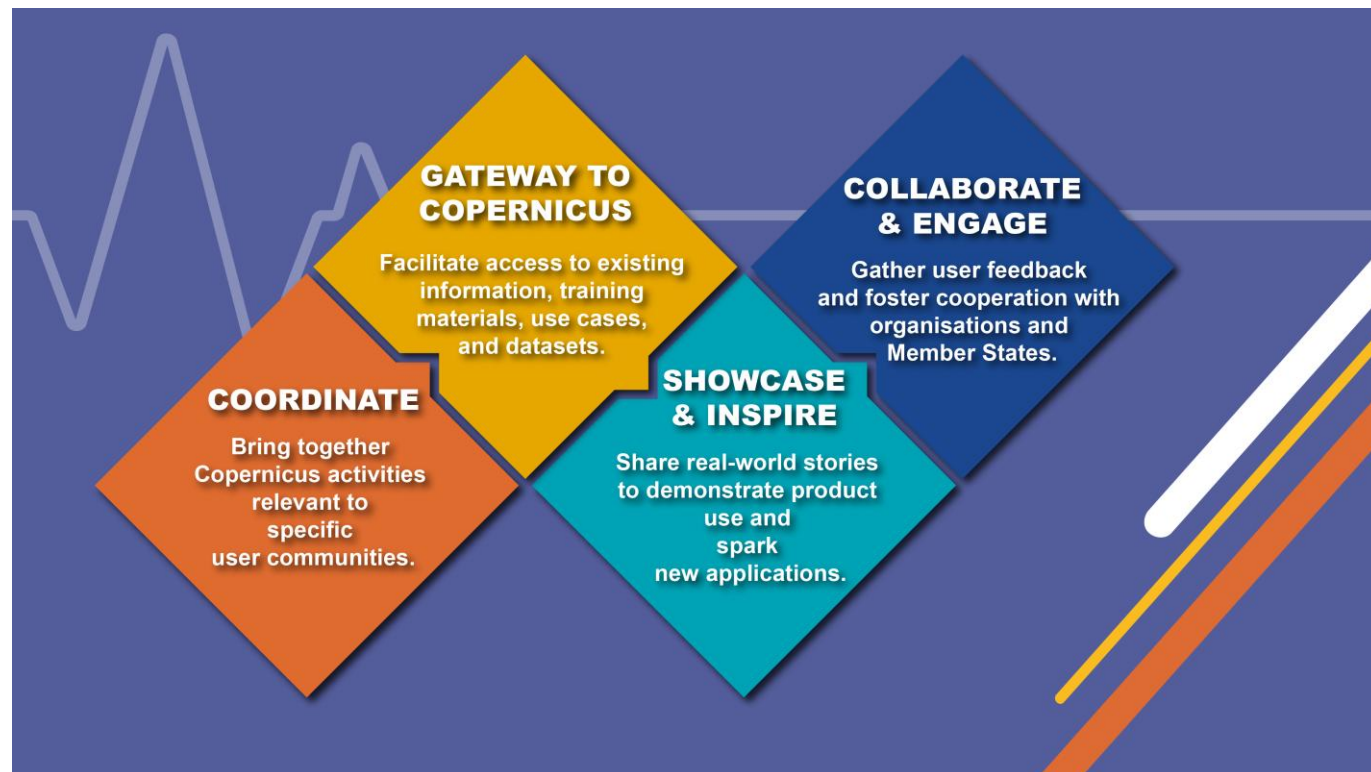
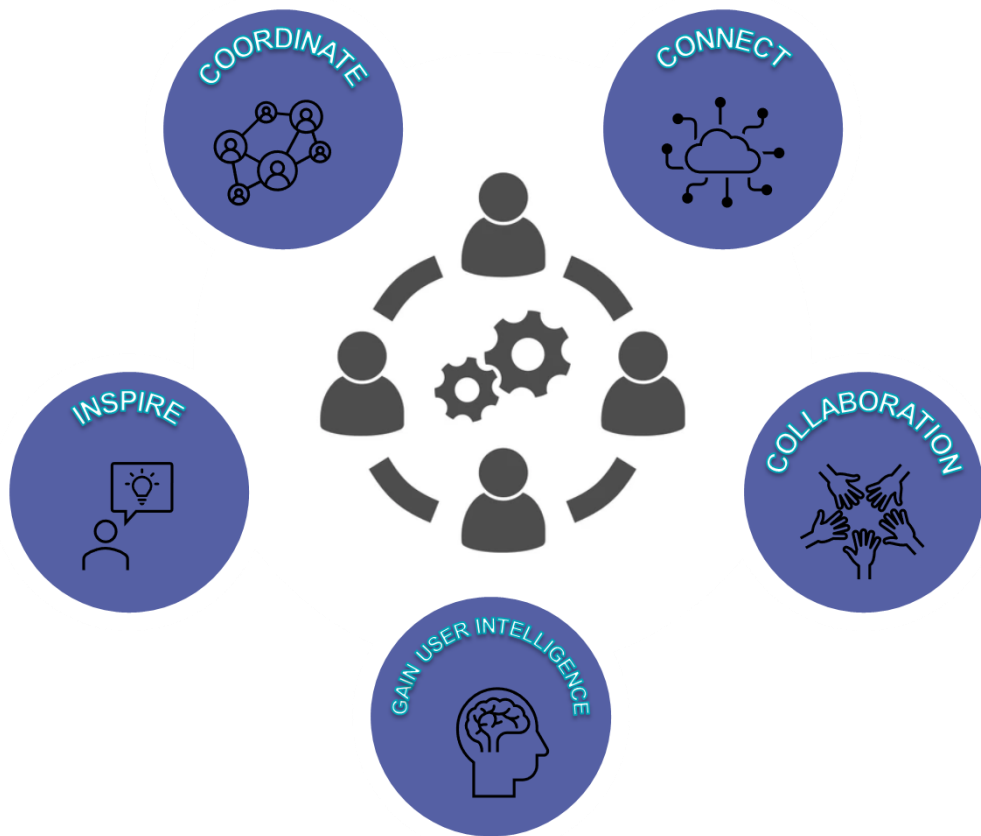
2nd Climate-Sensitive Vector Dynamics Modelling Workshop

17-19 September, Nicosia, Cyprus

Dr. Jonilda Kushta, Dr. Cristina Ananasso
ECMWF



The Copernicus Health Hub



<https://health.hub.copernicus.eu>



The Copernicus Health Hub key questions



What is the Copernicus Health Hub :

- A comprehensive platform that provides user-centered, targeted information on environmental factors affecting citizens' physical, mental, and social well-being.
- It aims to constitute an inter-disciplinary collaborative meeting point, where health and environmental science communities come together through the power of Copernicus, under the lead of CAMS.
- The Copernicus Health Hub embraces the definition of the World Health Organisation (WHO):
"Health is a state of complete physical, mental, and social wellbeing and not merely the absence of disease or infirmity."



Physical health



+

Mental health



+

Wellbeing





The Copernicus Health Hub key questions

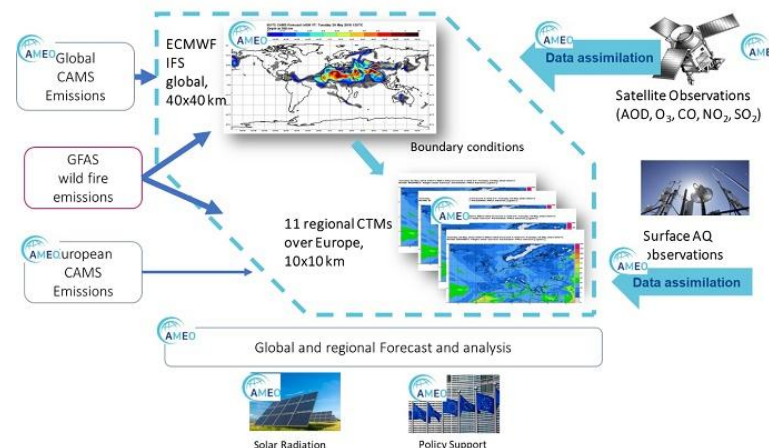


Why was it created:

Copernicus represents the perfect initiative of user driven approach leveraging EO based products.

Translating this wealth of data into actionable intelligence for relevant health communities, remains a critical challenge.

CHH is working on bridging this gap and bringing together the health players to connect, share, inspire and collaborate.





The Copernicus Health Hub key questions



How does it contribute:

- highlights cutting-edge products from the Copernicus ecosystem pertinent to health,
- showcases user stories of successful implementation while helping to build an active community
- integrates community feedback on health application needs.

PRODUCTS AND DATA

Access to free and open health-related products and data from the Copernicus Services

USE CASES

Use cases and articles showcasing how Copernicus Service data is applied in health-related fields

TRAINING AND EVENTS

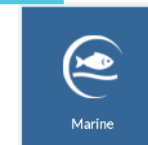
Health-related workshops, seminars, and events from across the Copernicus Services and other partner organisations



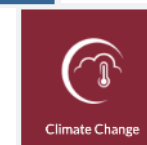
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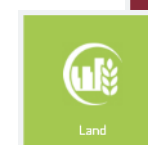
CAMS



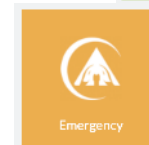
CMEMS



C3S



CLMS



CEMS



The Copernicus Health Hub key questions



Who can contribute to or benefit from CHH:

1. EU institutions and agencies
2. Researchers and universities
3. Industry and innovators
4. Policy makers and public authorities
5. The media and general public
6. Civil society organizations and NGOs



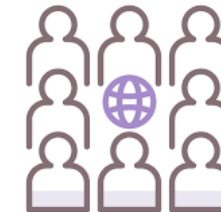
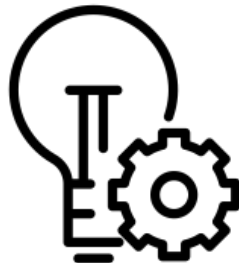
European
Climate and Health
Observatory



European
Environment
Agency



ESA EO4Health





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COPERNICUS HEALTH HUB

A comprehensive resource
of user-centered, targeted information
on environmental factors affecting citizens'
physical, mental and social well-being.



GLOBAL ATMOSPHERIC COMPOSITION

- KEY GASEOUS AND AEROSOL SPECIES
- OZONE LAYER
- UV INDEX

EUROPEAN REGIONAL AIR QUALITY

- KEY POLLUTANTS PM_{2.5}, PM₁₀, O₃, NO_x
- ALLERGEN POLLEN SPECIES

POLICY TOOLS

- CONTRIBUTION OF ANTHROPOGENIC AND NATURAL (e.g.dust) SOURCES
- PM₁₀ CHEMICAL COMPOSITION

EMISSIONS

- ANTHROPOGENIC SOURCES BY SECTOR
- FIRE EMISSIONS



Climate
Change

THERMAL COMFORT INDICES

- MEAN RADIANT TEMPERATURE
- UNIVERSAL THERMAL CLIMATE INDEX

CLIMATE VARIABLES

- AIR TEMPERATURE
- PRECIPITATION
- HUMIDITY
- WIND

CLIMATE EXTREME INDICATORS

- HEAT AND COLD EXTREMES
- RIVER FLOOD AND DISCHARGE
- FIRE DANGER INDICATORS
- SEA LEVEL RISE AND EXTREMES
- VECTOR HABITAT SUITABILITY INDEX



Emergency
Management

EARLY WARNING AND MONITORING

- FLOOD DATA
- WILDFIRE DATA
- DROUGHT-RELATED INDICATORS

EXPOSURE MAPPING

- SETTLEMENTS
- POPULATION

ON DEMAND MAPPING

- EMERGENCY
- PREPAREDNESS AND RECOVERY

(Selected near-real time data is restricted)



Marine
Monitoring

OCEAN BIOGEOCHEMISTRY

- CHLOROPHYLL
- NITRATE
- PHOSPHATE
- pH
- DISSOLVED OXYGEN

OCEAN PHYSICS

- SEA LEVEL HEIGHT AND TEMPERATURE
- SEA SURFACE WINDS
- WAVE HEIGHT AND INCIDENCE

MARINE HEAT WAVES



Land
Monitoring

LAND USE / LAND COVER MAPPING

BIO-GEOPHYSICAL PARAMETERS

- LAND SURFACE TEMPERATURE
- SOIL MOISTURE
- VEGETATION STATE
- IMPERVIOUSNESS

GROUND MOTION PRODUCTS

STREET-LEVEL GREEN SPACES

BURNT AREAS

LAKE WATER LEVEL AND QUALITY



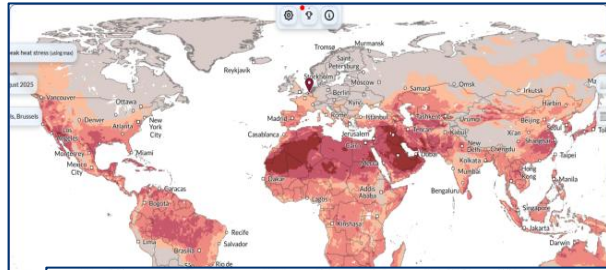
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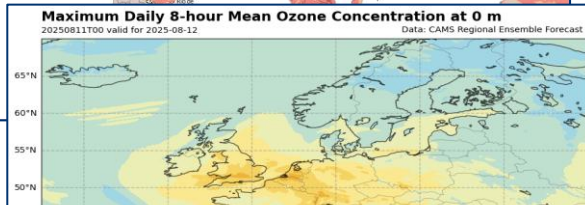
HEALTH HUB



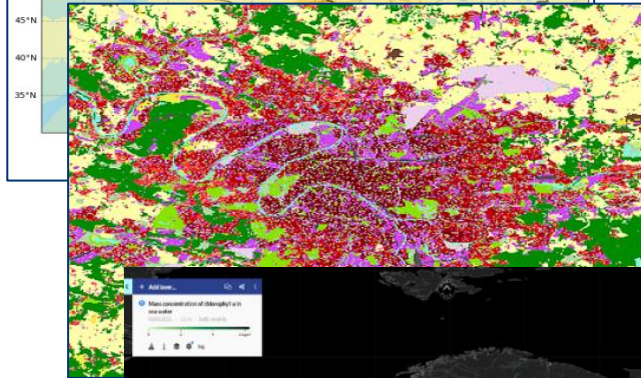
Examples of datasets and applications



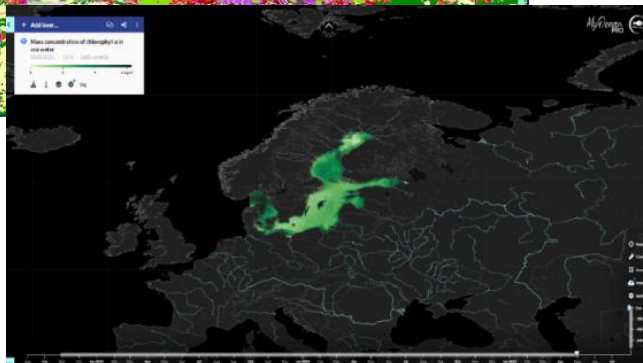
Thermal Trace App: Thermal stress, both from heatwaves and cold snaps, poses significant threats to human health, increasing risks of heat-related illnesses, cardiovascular distress, and cold-induced harm.



CAMS European air quality forecasts: This dataset is useful for understanding health impacts of air pollution, allowing researchers to analyse the link between air pollution and NCDs.



Street Tree Layer: This dataset supports research into the links between urban greenery and health outcomes such as reduced stress, improved cardiovascular health, and lower heat-related illness rates.



Baltic Sea Biogeochemistry Analysis and Forecast: This dataset supports the analysis of marine environmental quality and its potential impact on health, particularly concerning waterborne diseases and harmful algal blooms linked to eutrophication, and is useful for the prevention of bathing-related illnesses.



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Atmosphere
Monitoring Service
atmosphere.copernicus.eu

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Success stories and articles

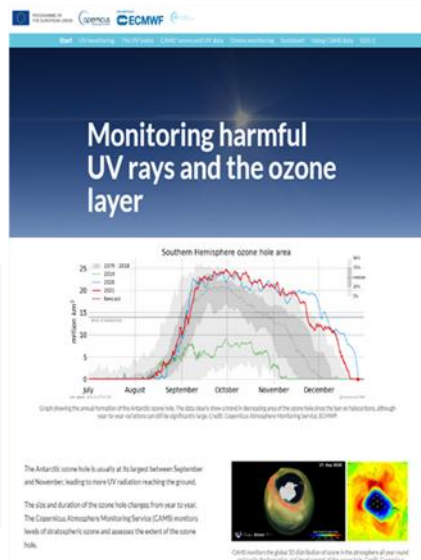


Intensification of climate extremes

Projections show that extreme temperatures will increase even more rapidly than average temperatures in the Mediterranean region. For instance, maximum temperatures on summer days may increase by up to 7°C by the end of the 21st century compared to the recent past. Heatwaves provide extra physiological stress on the human body and cause excess mortality.

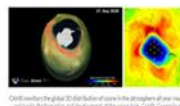
Urban centers are prone to the **heat island effect**, in which heat is trapped by narrow streets and pavements and human activities generate extra heat. Temperatures rise faster and reach higher extremes than in surrounding areas.

"We are finally starting to focus on heat waves, especially in urban"



The Antarctic ozone hole is usually at its largest between September and November, leading to more UV radiation reaching the ground.

The size and duration of the ozone hole changes from year to year. The Copernicus Atmosphere Monitoring Service (CAMS) monitors levels of atmospheric ozone and assesses the extent of the ozone hole.



CAMS monitors the global 3D distribution of ozone in the atmosphere of our world, with a focus on the Southern Hemisphere and the Antarctic ozone hole.



Heat stress in European cities

About three quarters of the European population live in urban areas, a proportion that is expected to grow in the coming decades. Urban environments are especially impacted by extreme temperatures, with air temperatures in European cities rising by 2°C to 3°C by the end of the 21st century. This is due to the urban heat island effect, where buildings and paved surfaces trap heat, and human activities add heat to the surroundings.

This urban heat island effect means that urban populations are more at risk than rural ones from the harmful effects of rising temperatures. Extreme heat can bring a range of health problems, and heat-related mortality is increasing. The impacts are especially dangerous for vulnerable ageing European populations.

The Finnish Institute of Technological Research (VTT) and partners used Copernicus Climate Change Service (C3S) data combined with ge



Forecasting pollen to alleviate allergy suffering

Using CAMS data to track and forecast pollen concentration across Europe

For many people, the arrival of spring is a time of celebration. The weather warms, daylight hours increase and plants begin to bloom.

For those flowering plants, however, it's a time of suffering. As pollen grains are spread through the air, causing itchy eyes, runny noses and sneezing, some people also experience more serious symptoms, such as shortness of breath or swelling of the airways.

Estimates suggest that one in four people suffer from pollen-related allergies. The costs associated with treatment and missed workdays from severe reactions can add up to around €200 billion each year.

The spread of pollen is directly related to weather conditions, offering an opportunity to use meteorological data and models to predict the pollen season. One of the key functions of the Copernicus Atmosphere Monitoring Service (CAMS) is to monitor and forecast air quality and pollen levels across Europe. Through the creation of a dataset that maps air quality and pollen levels across the continent, CAMS has enabled the development of several tools



Share your stories on how
you combine Copernicus data
and health information!

more



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Data access explanatory steps

 <https://health.hub.copernicus.eu>



**COPERNICUS
HEALTH HUB**
The link between Earth Observation,
environmental information, and health

IN A NUTSHELL



The environment we live in...
The environment we live in...
The environment we live in...

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Services

Search datasets Sector Region Data provider

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Search datasets Sector Region C3S Reset

Heat waves and cold spells in Europe derived from climate projections

This dataset is valuable for public health planning in the context of climate change, particularly for assessing how the increase in the incidence of heat waves and cold spells will affect the health of the population.



Search datasets Sector Region CLMS Reset

Impacts of climate change on human health

This dataset is valuable for public health planning in the context of climate change, particularly for assessing how the increase in the incidence of heat waves and cold spells will affect the health of the population.

Sector: Physical health, Environmental quality conditions, Epidemiology Region: Europe Data provider: CAMS

CAMS European air quality forecasts

This dataset is useful for understanding the short-term health impacts of air pollution, allowing researchers to analyse the link between air pollution and respiratory or cardiovascular diseases. It supports preventive public health interventions by providing air quality warnings, enabling authorities to implement mitigation strategies such as reducing outdoor activities or advising vulnerable populations during periods of poor air quality to reduce exposure to harmful pollutants. It provides hourly air quality analyses, updated daily, and forecasts for key pollutants such as nitrogen monoxide (NO), nitrogen dioxide (NO₂), sulphur dioxide (SO₂), ozone (O₃), particulate matter (PM_{2.5} and PM₁₀), dust and allergen pollen species over Europe. These forecasts are based on an ensemble of eleven air quality forecasting systems and are regularly validated against in situ ground-based observations, ensuring their reliability for health-related applications.

Sector: Physical health, Environmental quality conditions, Epidemiology Region: Europe Data provider: CAMS

CAMS European air quality reanalyses

This dataset can be used to study the long-term health impacts of air pollution, supporting research that links prolonged exposure to pollutants with chronic conditions such as lung cancer, cardiovascular disease, and neurodevelopmental disorders. It informs health policy by providing consistent historical data for exposure assessments. It includes main gaseous pollutants like nitrogen monoxide (NO), nitrogen dioxide (NO₂), sulphur dioxide (SO₂), ozone (O₃) and others, as well as particulate matter (PM_{2.5} and PM₁₀ and sub-components), and PM₁₀ from dust, sea salt and wildfires. The dataset integrates atmospheric modelling with ground-based observations to offer a reliable, long-term record of air quality. It covers Europe from 2013 to 2023, with hourly data available at multiple vertical levels in the atmosphere.

Sector: Physical health, Environmental quality conditions, Epidemiology Region: Europe Data provider: CAMS

Ozone Layer Monitoring

The ozone layer shields life on Earth from harmful ultraviolet-B (UVB) radiation, and its depletion has serious health consequences. Increased UVB exposure raises the risk of skin cancers, including melanoma, contributes to cataracts and vision impairment, and can weaken the immune system, making people more vulnerable to infections and disease. The Copernicus Atmosphere Monitoring Service (CAMS) tracks the state of the ozone layer by providing daily global information on ozone concentrations, with data records starting in 1979. Since 2015, CAMS has also delivered operational five-day forecasts, allowing authorities and health services to anticipate periods of elevated UV exposure.

European Air Quality Forecasts Optimised at Observation Sites

Air pollution is one of the most serious environmental threats to health, linked to asthma, chronic lung disease, cardiovascular problems, and premature deaths. The Copernicus Atmosphere Monitoring Service (CAMS) provides daily air quality forecasts at observation sites across Europe, helping health professionals and authorities anticipate harmful exposure levels. The dataset includes both raw forecasts from the European air quality model and refined analyses that use machine learning to adjust errors and improve local accuracy. It focuses on the pollutants most damaging to health, such as ozone (O₃), nitrogen dioxide (NO₂), and fine particulate matter (PM_{2.5} and PM₁₀). This supports early health warnings, preparedness of healthcare services, and evidence-based policies to reduce the health burden of air pollution across Europe.

Sector: Physical health, Well-being, Environmental quality conditions Region: Europe Data provider: CAMS

Global Biomass Burning Emissions (GFAS)

Wildfires and biomass burning release large amounts of smoke and pollutants that significantly affect human health. Exposure to fine particulate matter and toxic gases from these events is linked to respiratory and cardiovascular diseases, increased hospital admissions, and premature deaths. The Copernicus Atmosphere Monitoring Service (CAMS) provides global daily estimates of emissions from vegetation fires through its Global Fire Assimilation System (GFAS). Using satellite observations of fire intensity, GFAS delivers information on burned biomass and the release of harmful pollutants, including particulate matter and greenhouse gases. The dataset provides global coverage from 2003 to the present, updated daily. By quantifying the emissions from wildfires, GFAS supports exposure assessments, health impact studies, and public health preparedness during severe fire events that threaten air quality across continents.

Sector: Physical health, Well-being, Environmental quality conditions Region: Global Data provider: CAMS



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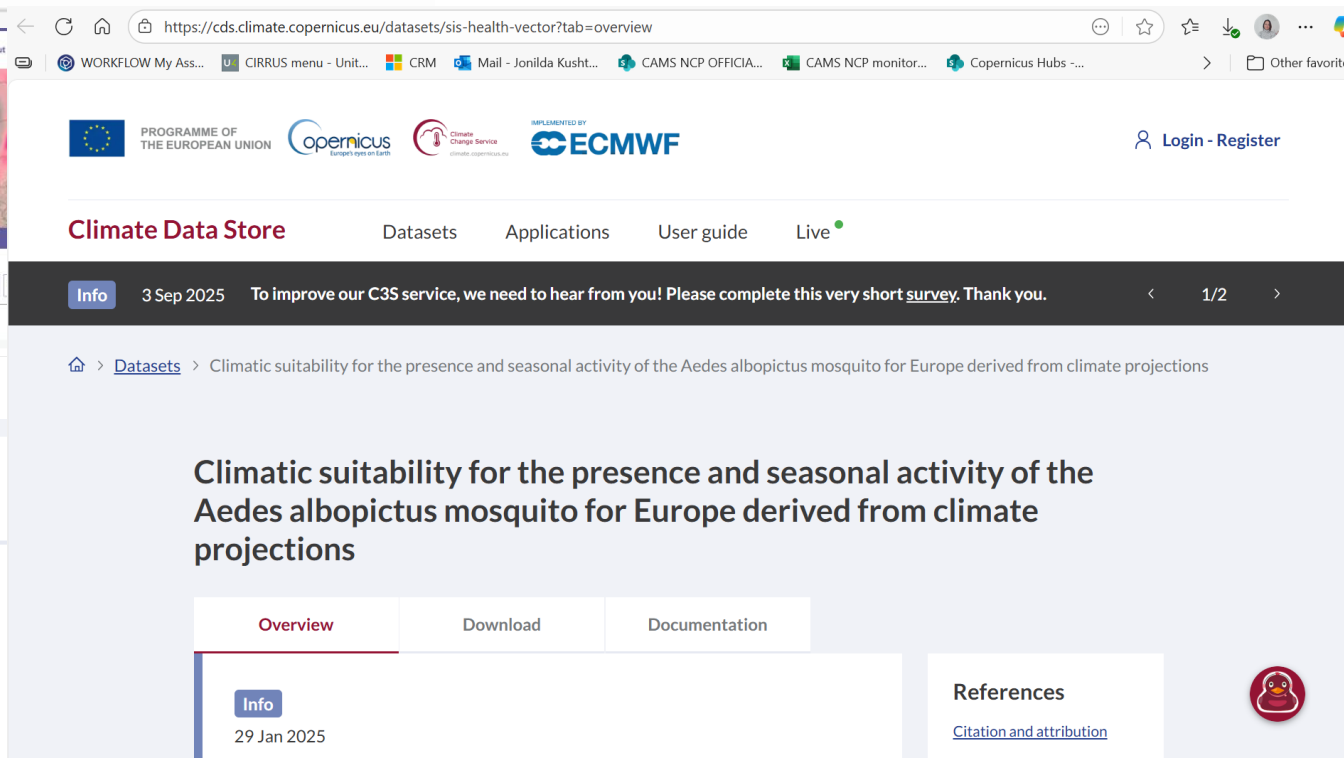


Data access explanatory steps

 <https://health.hub.copernicus.eu>



 <https://health.hub.copernicus.eu/products-and-data>



Overview

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Documentation



Overview

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29 Jan 2025

References

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Data access explanatory steps

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HEALTH HUB**
The link between Earth Observation,
environmental information, and health

<https://health.hub.copernicus.eu/products-and-data>

IN A NUTSHELL



The environment we live
in is not just our
radiation are not just our
right tools, environment
research, and policy.

**PRODUCTS AND
DATA**

Access to free and open health-related
products and data from the Copernicus

Climate Data Store Datasets Applications User guide Live

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You must accept the terms before submitting this request.

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API request

Please go to the [documentation page](#) for information as to how to use the CDS API.

Show API request code

Login/register to submit request

Request
validation

Request size

[Variable](#)

[Definition](#)

[Experiment](#)

[Ensemble statistic](#)

Climate Data Store Datasets Applications User guide Live

Complete all required fields before submitting the request.

Variable [Select all](#)

At least one selection must be made

☐ Heat wave days

☐ Cold spell days

Definition

At least one selection must be made

☐ Climatological
related

☐ Health related

☐ Country related

Experiment [Select all](#)

At least one selection must be made

☐ RCP4.5

☐ RCP8.5

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Publication date

2019-11-07

Update date

2025-01-31

Standard
metadata

[STAC](#)

[CSW](#)

Atmosphere Data Store Datasets User guide Live

Variable [Select all](#)

At least one selection must be made

☐ Alder pollen

☐ Ammonia

☐ Birch pollen

☐ Carbon monoxide

☐ Formaldehyde

☐ Glyoxal

☐ Grass pollen

☐ Mugwort pollen

☐ Nitrogen dioxide

☐ Nitrogen monoxide

☐ Non-methane volatile
organic compounds
(VOCs)

☐ Olive pollen

☐ Ozone

☐ Particulate matter < 2.5
µm (PM2.5)

☐ PM2.5, ammonium

☐ PM2.5, nitrate

☐ PM2.5, residential
elementary carbon

☐ PM2.5, secondary
inorganic aerosol

☐ PM2.5, sulphate

☐ PM2.5, total elementary
carbon

☐ PM2.5, total organic
matter

☐ Particulate matter < 10
µm (PM10)

☐ PM10, dust

☐ PM10, sea salt (dry)

☐ PM10, wildfires

☐ Peroxyacyl nitrates

☐ Ragweed pollen

☐ Sulphur dioxide

[Set new](#)

References

[Citation and attribution](#)

DOI: [10.24381/a4005cee](#)

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